Operations Research for Health Care 2 (2013) 66–74

Contents lists available at ScienceDirect

Operations Research for Health Care

journal homepage: www.elsevier.com/locate/orhc

Modeling the impact of changing patient flow processes in an emergency department: Insights from a computer simulation study

Renata Konrad [a,*], Kristine DeSotto [b], Allison Grocela [a], Patrick McAuley [a], Justin Wang [a], Jill Lyons [c], Michael Bruin [c]

[a] School of Business, Worcester Polytechnic Institute, Worcester, MA, 01720, USA

[b] New England Veterans Engineering Resource Center, 150 S. Huntington, Boston, MA, USA

[c] Emergency Department, St. Vincent Hospital, Worcester, MA, 01720, USA

ARTICLE INFO

Article history:
Received 2 July 2012
Accepted 23 April 2013
Available online 9 May 2013

Keywords:
Emergency department
Split-flow
Discrete-event-simulation
Resource allocation

ABSTRACT

We report on the use of discrete-event simulation modeling to support process improvements in a hospital emergency department (ED), namely the implementation of a split-flow process. Our partner hospital was effective in treating patients, but wait time and congestion in the ED created patient dissatisfaction, unsafe conditions and staff morale issues. The split-flow concept is an emerging approach to manage ED processes by splitting patient flow according to patient acuity and enabling parallel processing. We contrast the split-flow operational model to other types of ED triage. While early implementations of the split-flow concept have demonstrated significant improvements in patient wait times, a systematic evaluation of operational configurations is lacking.

We created a discrete-event simulation model and established its face validity for Saint Vincent Hospital in Worcester, USA, a community-teaching, Level II Trauma Center. Seventeen scenarios were tested to estimate the likely impact of a split-flow process redesign, including staffing level changes and patient volume changes. The scenarios were compared in terms of Door-to-Doctor time and length-of-stay for different patient acuity levels.

Findings from the study supported implementation of the split-flow improvements. Statistical analysis of data taken before and after the implementation indicate that waiting time measures were significantly improved and overall patient length-of-stay was reduced. To gauge the success of the current split-flow process at Saint Vincent we compare performance metrics from three different sources: benchmark metrics, hospital data prior to split-flow implementation, and performance metrics post implementation.

1. Introduction

Hospital-based emergency care is critically important to the health of a population [1]. Not only do emergency departments (EDs) provide urgent care, but they increasingly serve as adjuncts to community physician practices, especially in the United States [1,2]. Worldwide, ED visits are steadily increasing, with the United States reporting an annual increase of 3% [3]. Factors contributing to the growth include an aging population (3), limited access to medical care from other sources [4,5], and (at least in the United States) a rising trend towards utilizing the ED for non-emergency care [4–7]. As a result, the emergency department has become the main point of entry into hospitals and accounts for more than half of all admissions to most hospitals [7,8].

The surge in patient volume is a significant contributor to an international phenomenon known as ED crowding [9,10]. More than two thirds of US hospitals are affected by crowding [11], a situation when the need for emergency services outweighs available ED resources [12]. Countries around the world face similar circumstances [13–15]. Much has been published in both academic and lay literature about the negative effects of ED crowding. Excessive patient overload leads to medical errors and poor outcomes [14,16–19], prolonged patient wait times [20], increased patient complaints [21,22], decreased staff satisfaction [20], and decreased physician productivity [12,23,24]. Recent studies suggest that as the average length-of-stay of ED patients increases, the risk of death or hospital readmission within the next 7 days increases for those who were discharged or left without being seen [25,26]. In the United States, studies document the loss of revenue resulting from patient diversion and elopement [27,28].

In this paper we evaluate and guide one hospital's approach to emergency department crowding by implementing a split-flow process. At the beginning of the study, waiting times and patient

* Corresponding author. Tel.: +1 508 831 5192; fax: +1 508 831 5720.
E-mail address: rkonrad@wpi.edu (R. Konrad).

2211-6923/$ – see front matter © 2013 Elsevier Ltd. All rights reserved.
http://dx.doi.org/10.1016/j.orhc.2013.04.001

congestion at the emergency department were substantial. The average time patients spent at the ED was almost four hours, with the majority of that time spent waiting. The hospital planned to implement a split-flow process as an innovative approach to decrease ED patient length-of-stay and improve care quality. The central tenet of split-flow is the sooner patients can enter the hospital system, the sooner they are able to be treated and released. By physically splitting patient flow according to acuity and enabling parallel processing, high demand resources are conserved for higher acuity patients and throughput for all patients is increased. We first define a split-flow process for the reader and then evaluate various configurations for this process through a simulation model. Using our recommendations, the hospital selected an alternative prior to implementation and we present the improvements in performance metrics post implementation.

1.1. Research objectives

The objective of this paper is to evaluate the impact on patient throughput arising from different split-flow configurations. Our method enables a hospital to quantify the effects of system redesign prior to implementation and to examine how the split-flow concept can be best applied to their particular hospital, ultimately decreasing implementation costs and disruptions. We use simulation modeling as a mechanism to help in our diagnostic and improvement efforts in creating a decision-tool for a community-based ED in central Massachusetts, USA.

A contribution of our work is to show how a structured analysis of patient flow and an evaluation of a split-flow process can significantly improve waiting time and patient congestion in an ED. A simulation model can address concerns regarding resource allocation that may hamper implementation of the split-flow concept. We consider the impact of various resource strategies on two streams of patients to ensure that improvement for one type of patient is not at the expense of the other. A second major contribution is to demonstrate how simulation modeling and the quantitative analysis it provides can assist in convincing involved parties to implement improvements. Although multiple institutions have recently considered modifications to their triage and patient intake processes to improve throughput and crowding, published data on the effect of these interventions is unfortunately sparse [29,30]. Finally we fill a current void in ED implementation research and decision support for deploying a split-flow process.

1.2. ED triage and patient flow

Triage systems, a way to prioritize patient needs, provide an effective tool for departmental organization, monitoring and evaluation [31]. The formal concept of triage was introduced in emergency departments in the late 1950s [32], and sorting patients is now critical to the effective management of modern emergency departments [17,31]. Worldwide, hospitals split patient flow by acuity and/or by function in an effort to decrease wait times and promote quality. [8] reviews triage-related interventions to improve ED wait times in the United States and Europe. The authors evaluate five processes: streaming, fast-track, team triage, point-of-care testing, and nurse-requested X-ray. Patient streaming and the use of fast-tracks is not a new concept as EDs in many countries have used these techniques since the 1980's [6] (see [8,33] for recent reviews). However, the pervasiveness of ED congestion resulting from access block and increased demand has led to calls for a review of systems of triage [31]. The split-flow approach is considered a "new generation of EDs" in the United States and incorporates concepts from streaming, fast-track, and team triage. Split-flow literature is fairly scarce as it is a relatively new concept. Table 1 illustrates how the split-flow approach differs from a traditional emergency department without some sort of patient streaming, and the widely implemented fast-track system.

Several key differences make the split-flow process innovative and unique. First, the split-flow process delivers patients to resources and providers, as opposed to having these resources brought to the patient. Low acuity patients are directed to move through various areas of the split-flow configuration. Acuity is defined by the standard five-level Emergency Severity Index (ESI) used in the United States. The lowest acuity patients are assigned an ESI Level 5, while the most severely acute patients are assigned a Level of 1. In the split-flow model, after the initial intake, patients move to a "continuing care area" consisting of chairs and recliners to receive treatment. If appropriate, patients then move to a "results pending" area. Conversely in fast-track approaches, the patient remains in one bed (or in some cases a recliner) while resources are brought to them; providers come to the patient, results are delivered to the patient and so on. A second key difference is bed occupancy. In the split-flow model, the continuing care area eliminates bed occupancy for lower acuity patients while some fast-track implementations keep patients in a bed in a designated part of the ED. A third key difference between split-flow and other ED operational models is provider involvement. In a split-flow model, a dedicated physician sees lower-acuity patients, ideally in a team with a nurse and technician. In contrast, lower acuity patients in fast-track models are typically seen by mid-level providers such as nurse practitioners—who are clinical professionals with advanced registered nursing qualifications. Finally, traditional and fast-track EDs have a sequential diagnostic process, which adds to the overall ED stay. Split-flow uses a parallel process for diagnostic testing (as do some other approaches, including physician-directed queuing [35]; team triage [36], and rapid entry and accelerated care at triage (REACT) [37]) (see [6] for a further discussion).

We further describe a generic split-flow process in Fig. 1. The process routes (splits) lower acuity patients as defined by the standard five level Emergency Severity Index (ESI) [38] in a separate queue from those higher acuity patients who are awaiting placement in a traditional ED bed. Lower acuity patients are designated by an ESI of 5, 4, or sometimes 3, while higher acuity patients have an ESI of 1, 2, or sometimes 3. By not placing lower acuity patients in beds, limited bed capacity is conserved for higher acuity patients requiring a bed immediately. By offering a different treatment model to lower acuity patients, EDs expect to reduce bed occupancy and increase their overall capacity. As the split-flow approach is still in its infancy, few studies are able to validate this claim.

Challenges in implementing a split-flow process, or similar types of models are anecdotal and have not been studied much to date. Our contrition is to demonstrate that the split-flow concept yields improved performance metrics when compared to a traditional ED model. Further, we recognize that determining an appropriate resource allocation for a split-flow implementation is a challenge for hospitals. The results of this paper also provide a contribution to the literature in this respect.

1.3. Healthcare simulation

Healthcare simulation is an active area of research and practice, with the ED being one of the most popular areas for modeling. [39] suggest several reasons for this popularity. Most EDs are relatively self-contained with easily observable processes over relatively short time periods. Improvements in performance are easier to demonstrate and link to specific actions, which may not be true elsewhere in healthcare. [40] reviews simulation studies which investigate ED congestion while [41] reviews a number

Table 1
Comparison of four emergency department operational models.

	Traditional ED	Traditional fast-track	Low acuity fast track	Split-flow
Patients streams	None; all patients move through the same process flow	ESI levels 4, 5 and some ESI level 3 are separated from ESI level 1, 2 and some 3	Only ESI level 4 and 5 are separated from ESI level 1, 2 and 3	ESI levels 4, 5 and some ESI level 3 are separated from ESI level 1, 2 and some 3
Triage function	Spot check or comprehensive triage	By function and/or acuity	By function and/or acuity	"Quick look" triage and/or team triage, by acuity
Resources	All patients occupy a bed; Physicians, nursing staff and mid-level provider involvement	All patients occupy a bed; Lower acuity patients are seen by mid-level provider involvement (Physicians Assistants, Nurse Practitioners) and occasional physician involvement	Lower acuity patients use a treatment area (may or may not contain beds) and share a room; Physician Assistants and nursing staff involvement	Lower acuity patients go to "continuous care area" with chairs; reduced bed occupancy; physicians, nursing staff and mid-level provider involvement
		Higher acuity patients follow traditional ED model	Higher acuity patients follow traditional ED model	Higher acuity patients follow traditional ED model
Diagnostic process	Sequential	Sequential	Typically parallel	Parallel
Physical layout	All patients in one area	Dedicated space, typically adjoining the ED	Dedicated space, typically adjoining the ED	Dedicated space, typically adjoining the ED
Advantages	Patients are all seen by a physician	Can dedicate mid-level providers to fast-track area	Fewer beds are required [34]	Conserves beds for appropriate patients, decreased time-to-physician assessment [34]
Disadvantages	Sequential care process for all patients creates backlog	Sequential care process duplicates main ED backlogs, space reconfiguration [34]	Low volumes of low-acuity patients may underutilize fast track resources [34]	Physicians are involved in treating low-acuity cases, space reconfiguration required [34]

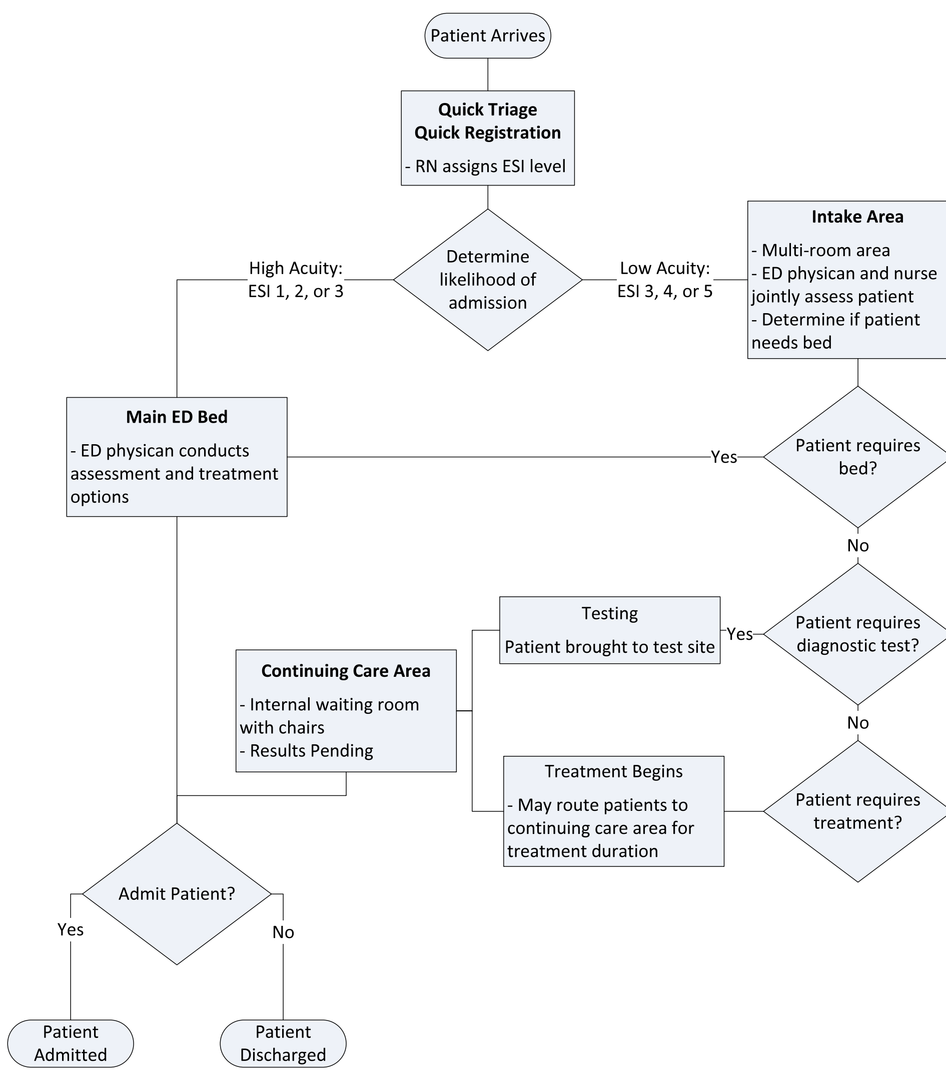

Fig. 1. Split-flow management process.

of analytical and simulation models aimed at addressing ED crowding.

The majority of ED simulation studies are motivated by costs, efficiency, re-engineering, and quality of service. Given these motivations to cut costs and increase efficiency, a primary goal of these studies is to examine causes of inefficiencies in processes and resource utilization. For example, studies examined patient flows [42] and bottlenecks in flows [43,44], causes of excessive wait times [42,44], and patient throughput [45,46]. A second major goal is to optimize resources such as staff and beds. Several studies examined alternative staff schedules [47–49], assessed the effect of different staff schedules on wait times [50,51], and quality of service [52] to recommend cost-effective schedules. Since beds are an important resource in the ED, studies examined critical bed requirements [53,54] and the impact of bed availability on wait times of admitted patients [55,56]. There are many examples of ED models that are built for specific hospitals such as [42,57], while others are intended to be reusable in other hospital settings [58–60].

A few simulation studies have also evaluated the effects of introducing fast care for low acuity patients [61], not providing care to low acuity patients [62], and redesigning processes to reduce patient length-of-stay (LOS) [63]. [64] used simulation to evaluate establishing a fast-track lane adjacent to the main ED and a large capacity lab in order to improve patient flow at the University of Louisville Hospital. [65] evaluated a similar layout when simulating the operation of the Children's Hospital of Eastern Ontario.

While numerous ED simulation studies appear in the literature, few recommendations are actually implemented. [66] offers some organizational insights as to why implementation rates are low, for example hospital management not being involved in the creation of the model. [67] found that only half of simulation models (of surgical patients) were constructed to address the needs of policy-makers, and only 26% reported some involvement of health system managers and policy-makers in the simulation study. [68] suggests that hospital management should be directly involved in the development of the simulation project in order to bolster the model's credibility, and should incorporate visual aids or animation to heighten users' confidence in the model's ability.

2. Methods

2.1. Setting description

Like many hospitals in the United States, Saint Vincent Hospital, located in Worcester, Massachusetts, is confronted by ED crowding, long wait times, and poor patient satisfaction. Saint Vincent Hospital, part of the Vanguard Health System, is a 348 bed acute care, community teaching, Level II Trauma Center hospital [30]. Saint Vincent serves not only the greater Worcester area, but also Worcester County at large with a population of 650,000. The ED is the largest department of the hospital, bringing in nearly half of all hospital admissions and this volume is growing. In 2012 over 68,800 ED patients were treated.

As with many hospitals in the United States, the Saint Vincent ED management team is struggling to decrease patient wait times, the time a patient must wait to see a physician, and total length-of-stay even with an increasing patient volume. ED staff is dedicated and keenly aware of the urgency of treating patients; however the hospital has been receiving below-average performance on patient satisfaction scores, particularly related to the metric of "waiting time to see a doctor" when compared to other hospitals in the Vanguard system. In response, Saint Vincent management decided to implement a split-flow process given early success stories in the industry. Sensitive to patient satisfaction and wishing to minimize organizational disruptions, the hospital proposed to assess the impact of a split-flow process on various performance metrics prior to implementation targeted for early 2012.

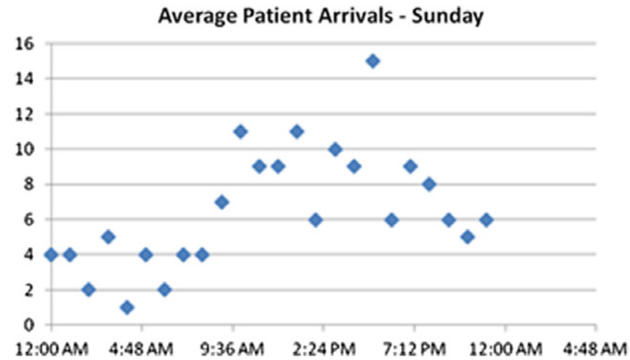

Fig. 2a. Patient arrival patterns Sunday.

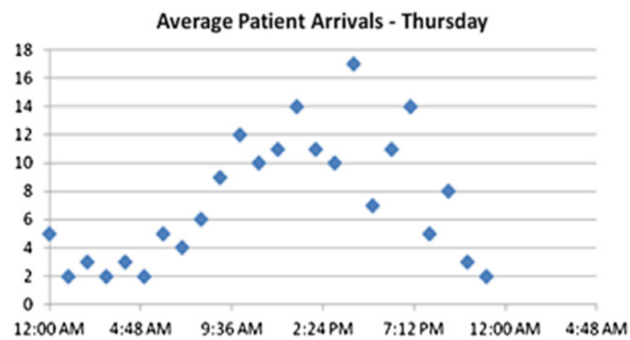

Fig. 2b. Patient arrival patterns Thursday.

2.2. Data collection and interpretation

Our data set is comprised of the entire population of patients who registered with the ED department during 2010, representing 63,828 unique visits. Each record consisted of time stamped events such as arrival, registration, discharge, transfer location and Emergency Severity Index (ESI). From this data, the following simulation inputs were calculated:

- Average daily patient arrivals by hour;
- Average number of patients admitted by day;
- Average number of patients discharged by day;
- Average number of patients transferred by day;
- Percentage of Emergency Severity Index (ESI) patients by day.

Arrival data was analyzed for variation in patient arrivals by season, month, week, day, and hour. Three variables—season, month, and day of week, did not impact arrival rates ($P > 0.05$). For example, Figs. 2a and 2b illustrate hourly arrival patterns for Thursday and Sunday. However, hourly inter arrival times were found to be statistically significant, and thus were used in the model.

Service times for patient evaluation by a physician, nursing care, EKG testing, lab work-up, triage and registration were obtained through observations and interviews with hospital staff. These times were differentiated according to high or low acuity patients, although we did not distinguish between ESI levels. Table 2 summarizes the data gathered and used in our model.

2.3. Performance measures

The objective of the simulation model was to understand the impact of alternative split-flow operational configurations on system performance. The ability to see and treat patients in a timely manner is important to hospital administrators who are focused on reducing patient wait times while providing high-quality treatment. The average length-of-stay (LOS) for all ED

Table 2
Simulation model inputs and outputs.

Inputs		Outputs
Design information	*Historic information*	*State information (performance metrics)*
(obtained from observations and interviews)	(extracted from MEDHOST)	(extracted from the simulation model)
• Control and data flow • Organizational model (roles, resources, etc.) • Service time distributions	• Patient arrival rates • Resource availability	• Total length-of-stay • Door-to-Doctor time

patients in a split-flow ED was deemed to be the most important objective by management. The LOS is defined as the earlier occurrence of registration or triage, to the time the patient physically leaves the ED. The secondary performance measure is Door-to-Doctor time (D2D), defined as the time from a patient's entrance into the system until their first encounter with a nurse, physician or physician's assistant. These performance metrics are listed in Table 2. Note that although the metric "patients leaving without being seen" is often used as an indicator of ED performance, it was excluded from the simulation model. Reasons why patients leave without being seen are complex, for instance physician factors, especially emergency medicine training [69], patient impatience during peak ED periods [70], correlation with the National Emergency Department Overcrowding Scale [71] among others. Such factors would require a number of modeling assumptions which would be beyond the scope of this analysis.

A key difference between traditional EDs and split-flow EDs is bed utilization. A split-flow configuration requires floor plan changes, which are typically subject to the structure of the hospital. Initially we wanted to examine bed allocation between patient types; however as this number was constrained by the hospital's floor plan, we would not be able to offer any solutions regarding how many beds should be in each area of the ED. Subject to structural constraints, the total number of beds in the ED was reduced from 38 to 30 and remains fixed.

2.4. Simulation construction

Several months of on-site observations and interviews with various medical and administrative staff established a thorough understanding of current and proposed ED processes. We refer the reader to the schematic in Fig. 1 which outlines the proposed split-flow design. The basic steps of the process include a "quick-look" triage, registration, bed allocation if appropriate, treatment and discharge. Upon arrival, a patient is triaged and assigned an ESI acuity level which determines whether the patient will follow the traditional route (high acuity) or the split route (low acuity). High acuity patients receive a bed that they "own" for the entirety of their stay. After an initial assessment, low acuity patients are sent to the continuous care area. The continuous care area consists of reclining chairs to treat patients. All patients will then either be admitted as an inpatient or discharged from the ED.

To create the simulation model, we duplicated the proposed ED environment in the discrete-event simulation software package ARENA, Version 12.0. Using the patient flow process descriptions and their corresponding activity flow for each patient acuity level, we translated process diagrams into ARENA simulation logic. Results from the simulation model were analyzed using the ARENA Output Analyzer.

Several assumptions simplified the modeling effort and eliminated insignificant parameters. It is assumed that each patient arrival corresponds to one person, not including family members or others who will not receive treatment. Further, low acuity patients have a dedicated physician for treatment in the low acuity treatment area. Another assumption concerns the assignment of nurses to specific areas providing additional support to administer diagnostic tests during periods of understaffing in the ED.

We ran the simulation model for one full week, replicated 140 times. To approximate the number of replications, the average half-width for all performance metrics was calculated after 5 initial runs and the number of runs was calculated such that the half-width of each confidence interval for a performance metric was no more than 5% of the average mean.

Table 3
Performance metrics for the current and simulated emergency department with 95% confidence intervals (based on 140 runs).

	Current ED performance (2010)	Simulated split-flow process (base model)
Door-to-Doctor (min)	59.0 ± 3.6	44.62 ± 0.79
Length-of-stay (min)	240.3 ± 6.5	130.56 ± 1.54

2.5. Model validation and verification

Throughout the design and development of the simulation model, several techniques were employed to validate the model including [72,73]:

1. Eliminating all error messages to ensure that entities are flowing through the model correctly.
2. Validating model logic and assumptions by reviewing the simulation model with clinical staff and management, including schedules.
3. Using quantitative techniques to test the model's assumptions: Input data analysis was validated by using goodness-of-fit tests as well as by graphical methods. A scenario analysis was also applied to measure the response of model performance results to changes in input parameters. The model was run under extreme conditions and results were analyzed, concluding that the model performed as expected under all conditions. Each condition was run for 140 replications.

3. Results

The data-driven simulation tool developed tested several scenarios arising from managerial and operational concerns. In particular, management was interested in how many physicians should be dedicated to different treatment areas, and how the system would perform should patient volumes or acuity distribution change.

As a benchmark, we first simulated a base scenario, in which resources and patient volumes were set at current levels. The base case is comprised of 30 ED beds, 9 recliners representing the "continuous care area", and an EKG area. Human resources in the base case were subject to schedules which varied by the time of day, but consisted of up to four physicians divided between the two processes, up to five residents, up to fourteen nurses between the two areas and intake, up to 2 administrative clerks, and up to 4 technicians. This is a realistic and feasible scenario, providing a good benchmark to evaluate alternative process configuration scenarios of interest to the hospital. Prior to evaluating alternative configurations, we first compare the base case, using 2010 data, against performance metrics within the emergency department for the same time period. Such a comparison was used to support the hospital's intuition that split-flow will indeed positively impact its measures of interest. The results of this comparison are found in Table 3. As expected, Door-to-Doctor and total length-of-stay significantly decreased. We note the drastic difference in the length of stay. Our simulation model focused on

Table 4
Performance metrics with confidence intervals for two resource allocation strategies (based on 140 runs).

	Base case		Scenario 1: add physician to low acuity area		Scenario 2: add physician to high acuity area	
	High acuity patients	Low acuity patients	High acuity patients	Low acuity patients	High acuity patients	Low acuity patients
Door-to-Doctor (min)	49.9 ± 0.2	23.4 ± 0.1	49.7 ± 0.1	23.2 ± 0.1	49.9 ± 0.1	23.3 ± 0.1
Length-of-stay (min)	139.0 ± 0.5	122.1 ± 2.2	140.2 ± 0.4	74.2 ± 0.6	137.2 ± 0.5	122.2 ± 2.3

Table 5
Performance metrics with confidence intervals for changes in patient arrival patterns (based on 140 runs).

	Base case		Scenario 3: impact on system performance with an increase in patient arrivals		Scenario 4: shift towards higher acuity patients		Scenario 5: shift towards lower acuity patients	
	High acuity patients	Low acuity patients	High acuity patients	Low acuity patients	High acuity patients	Low acuity patients	High acuity patients	Low acuity patients
Door-to-Doctor (min)	49.9 ± 0.2	23.4 ± 0.1	52.0 ± 0.7	24.1 ± 0.1	51.3 ± 0.2	23.2 ± 0.1	51.1 ± 0.04	21.66 ± 0.1
Length-of-stay (min)	139 ± 0.5	122.1 ± 2.2	151.7 ± 0.7	126.3 ± 2.7	146.3 ± 0.6	119.8 ± 2.3	140.0 ± 0.6	127.0 ± 3.0

the "front-end" operations, the processes involved in getting to the point in which the physician saw the patient. These processes were modeled in detail, and validated by management as discussed previously. Actual treatment processes were modeled with less detail. The purpose of the analysis was to use the model to comparative configurations prior to implementation of the split-flow approach.

Seventeen alternative split-flow configurations were assessed which examined the addition of human resources such as physicians, nurses, and technicians to the two processes; adding capacity such as an additional EKG room; and examining changes in patient volumes and acuity. Table 4 highlights the most significant configurations. We present the results by the two acuity streams of patients so that the reader can assess if the reported benefits of the split-flow process apply to both classes. We wanted to investigate whether the performance for lower acuity patients deteriorated at the expense of high acuity improvements (or vice-versa).

The first scenario we wish to emphasize added a dedicated physician to the split-flow area to help treat lower acuity patients. In the second scenario, an additional dedicated ED physician was assigned to treat higher acuity patients. These scenarios were based on the suggestions of [33] who recommend having a geographically dedicated area of the ED staffed by dedicated medical and nursing staff. Table 4 compares the performance metrics for these alternative configurations against the base model. The Door-to-Doctor time did not significantly change in either scenario, and the total length-of-stay did decrease significantly as a result of adding on a physician in the lower acuity treatment area (Scenario 1).

In addition to evaluating various resource configurations, our model examined the impact of anticipated changes in patient arrivals (Scenario 3) and the distribution of patient acuity (Scenarios 4 and 5). Although EDs across the United States are experiencing an average 3% annual increase in patient visits, we investigated the impact of a 2% annual increase over five years as this is reflective of demographic changes in Worcester County. Further, we tested how the three performance metrics would be affected should the distribution of patient acuity change. We first considered a scenario that increased the number of higher acuity patients (ESI-1 and ESI-2 patient levels increased by 20% and 10% respectively, and remaining levels adjusted accordingly). Similarly, lower acuity patients were increased (ESI-5 and ESI-4 patient levels increased by 20% and 10% respectively, while the remaining levels were adjusted accordingly). Table 5 summarizes the results of these scenarios.

As expected, the increase in patient arrivals significantly impacted performance metrics. As patient arrivals increased all performance metrics declined (Scenario 3). When the acuity level of patient arrivals shifted towards higher acuity patients, again, the results were not surprising. Greater volumes of high acuity patients generated longer wait times to see a provider and longer lengths of stays. In this scenario, the same performance metrics increased slightly for lower acuity patients reflecting increased demand for shared services such as labs, diagnostics and clerks. High acuity patients generate a greater amount of requests for lab and diagnostic services and in doing so decrease the capacity of these services for lower acuity patients. Similarly, performance metrics for both groups of patients declined when the patient acuity distribution shifted towards a greater number of lower acuity patients. Initially we were surprised by these findings; however upon investigation we found that the reasons for the difference are that the staffing levels and the physical layout are set for the current distribution of acuity. In this scenario, low acuity patients were queuing for recliners and resources, while the demand for EKG usage would also change across both classes of patients. Should the acuity distribution change significantly, Saint Vincent will have to reevaluate their resource allocation.

4. Discussion

Our primary objective was to develop a comprehensive data-driven simulation tool to illuminate the relationship between key system parameters in the ED, such as staffing levels and patient volume, against patient wait times for a split-flow process. We examined and evaluated various system configurations and policies simulated with data obtained from clinical and administrative information systems, as well as with manually collected data.

Prior to implementation, Saint Vincent reviewed the results of the study. Using the model, we illustrated that the most significant impact on their performance metrics could be achieved by adding a physician to the split-flow area. In such a configuration, the length of stay for lower acuity patients would be reduced by 48 min compared to the initial configuration Saint Vincent had proposed (from 122 to 74 min). Based on our recommendations, prior to the split-flow process being implemented, management added an additional Physician's Assistant to the lower acuity area. In early January 2012, the hospital implemented the split-flow process.

To gauge the success of the current split-flow process at Saint Vincent, in Table 6 we compare performance metrics from three different sources: (1) Saint Vincent data prior to split-flow implementation (calendar year 2010 and 2011), (2) Saint Vincent data after split-flow implementation (calendar year 2012), and (3) benchmark metrics from Vanguard. The results in Table 6 are

Table 6
Performance metrics before and after split-flow implementation, compared to benchmark.

	Benchmark	January 2010–December 2010	January 2011–December 2011	January 2012–December 2012
Door-to-Doctor (min)	30	59.0 ± 3.6	41.6 ± 8.9	31.1 ± 3.4
Length-of-stay (min)	270	240.3 ± 6.5	236.1 ± 13.7	220.6 ± 9.5
# LWBS/month	113	95.0 ± 13.8	91.8 ± 17.4	60.6 ± 13.4
Arrival to bed time (min)	15	9.7 ± 1.9	24.0 ± 4.2	14.6 ± 3.3
Monthly volume	5567	5304 ± 154.2	5411.4 ± 136.1	5734.0 ± 210.6

aggregated for the two types of acuity as prior to 2012 Saint Vincent did not have a method to track the two streams.

As the 2012 data shows, the split-flow process showed significant improvements in Door-to-Doctor time, total length-of-stay, arrival to bed time, and the number of patients left without being seen, even while the hospital experienced increases in patient volume of 6% between 2011 and 2012. These improvements are impressive and demonstrate the success of split-flow at Saint Vincent. Much of this success can be attributed to the hospital's implementation of the process and their inclusion of staff member input in all changes.

During our study, we concluded that one of the few negatives associated with split-flow implementation was the resistance of ED staff to change [74,75]. Some early adopters of split-flow did not initially see desired results because staff members did not fully buy into the process. Management at these hospitals failed to provide adequate information about the potential benefits of split-flow and the roles ED staff must play prior to its implementation [74,75].

As a result of these findings and the encouragement of Saint Vincent management, our team conducted a brief survey to gain ED staff feedback on the new split-flow process. The survey was aimed at answering the following questions such as: Do staff members think patient wait times are currently an issue? Do staff members feel additional changes still need to be made to the ED? Are staff members willing to change their roles and routines often to continuously improve ED performance? Are staff members well informed about the goals of split-flow?

The survey was conducted approximately one month after split-flow implementation at Saint Vincent. The survey results were encouraging as staff indicated that the majority considered themselves "very familiar" with split-flow principles and were willing to alter their roles to improve ED performance. The anonymous survey brought up several organization issues, which were classified as work assignment, and hospital-wide patient flow. For example, one staff member commented:

"(Split-flow) *Feels like more is being passed onto the nurse* (triage almost every patient now instead of just ambulance patients) and have to answer phone calls on all patients on team—so now we have to take more time explaining what is going on to another person. Feels *busier* and also feels like *mistakes are easier to make*—not a great feeling on some shifts".

A second response notes that:

"The *throughput time* from admission to inpatient unit *needs to be improved* thus decreasing ED wait times (there will be stretchers available). Split-flow working well upfront (former triage area)".

The hospital is currently addressing the above noted concerns by commissioning a study examining downstream bed availability and holding regular meetings with all nursing staff.

One limitation of this paper is the focus on a specific ED, so results may not be generalizable to other settings. Even in this case, the study fulfills an identified need by examining the split-flow process in a systematic way. It also applies improvement science to ED patient flow.

5. Conclusion

As hospitals seek to address lengthy, unsafe ED wait times, process redesign is often considered. In the United States, the split-flow concept is an emerging approach to manage ED processes by splitting patient flow according to patient acuity and creating parallel processes. Those patients who are less sick are split off from the traditional ED process flow, which is reserved for higher acuity patients. While early implementations of the split-flow concept have demonstrated significant improvement in patient wait times, a systematic evaluation of operational configurations is lacking. In this paper we build a discrete-event simulation model to evaluate various resource allocation strategies and examine the impact of realistic changes in patient arrival patterns.

Our simulated split-flow model showed statistically significant improvements the average length-of-stay, and Door-to-Doctor time which concur with anecdotes from early demonstration projects. When alternative resource allocation strategies were evaluated, the most significant improvement was the addition of a nurse or physician on the Door-to-Doctor time. The management of the study hospital responded to these finding by adding a Physician's Assistant to the low acuity area.

Saint Vincent has an active Quality Improvement and Lean Engineering program. While this program likely helped in creating a culture more accepting of change, the modeling we performed provided a systems perspective and quantitative evidence which supported the hypothesis that improvements do work.

We recognize that interactions between the ED and other departments and services in the hospital strongly affect the ED length-of-stay [76]. Studies have shown that interventions aimed at factors external to the ED have been successful in reducing ED crowding [77]. Few ED simulation studies exist which study the ED as part of a larger health system [40]. We are currently working with Saint Vincent to develop a more encompassing model.

References

[1] Institute of Medicine Committee on the Future of Emergency Care in the US Health System, Hospital-Based Emergency Care: At the Breaking Point, National Academies Press, 2006.
[2] C.J. Peek, Institute for clinical care improvement, report on the current state of the art for medical/health care home, Available at: http://www.health.state.mn.us/healthreform/homes/documents/HCH_ICSIReport.pdf (last accessed: June 2012).
[3] F. Zilm, J. Crane, K. Roche, New directions in emergency service operations and planning, J. Ambulatory Care Manage. 33 (4) (2010) 296–306.
[4] N.R. Hoot, L.I. LeBlanc, I. Jones, S.R. Levin, C. Zhou, C.S. Gadd, D. Aronsky, Forecasting emergency department crowding: a discrete event simulation, Ann. Emerg. Med. 52 (2) (2008) 116–125.
[5] N.R. Hoot, D. Aronsky, Systematic review of emergency department crowding: causes, effects, and solutions, Ann. Emerg. Med. 52 (2) (2008) 126–136.
[6] S. Welch, S. Davidson, Exploring new intake models for the emergency department, Am. J. Med. Qual. 25 (3) (2010) 172–180.
[7] P. Owens, A. Elixhauser, Hospital admissions that began in the emergency department, Agency for Healthcare Research and Quality, Healthcare Cost and Utilization Project, Statistical Brief #1, February 2006. Available at: www.hcupus.ahrq.gov/reports/statbriefs/sb1.jsp (last accessed: April 2012).
[8] S. Oredsson, H. Jonsson, J. Rognes, L. Lind, K.E. Göransson, A. Ehrenberg, K. Asplund, M. Castrén, N. Farrohknia, A systematic review of triage-related interventions to improve patient flow in emergency departments, Scand. J. Trauma Resusc. Emerg. Med. 19 (1) (2011) 43–52.
[9] D. Bertsimas, M.V. Bjarnadóttir, M.A. Kane, J.C. Kryder, R. Pandey, S. Vempala, G. Wang, Algorithmic prediction of health-care costs, Oper. Res. 56 (6) (2008) 1382–1392.

[10] R.J. Bittencourt, V.A. Hortale, A systematic review interventions to solve overcrowding in hospital emergency services, Cad. Saúde Pública. 25 (7) (2009) 1439–1454.
[11] J. Knapp, T. Bojko, M. Dolan, K. Frush, R. Furnival, D. Isaacman, S. Krug, R. Sapien, K. Shaw, P. Sirbaugh, Overcrowding crisis in our nation's emergency departments: is our safety net unraveling? Pediatrics 114 (3) (2004) 878–888.
[12] R.B. Case, D.L. Fite, S.M. Davis, S. Hoxhaj, W.P. Jaquis, T. Seay, C.S. Yeh, Emergency department crowding information paper. Irving (TX): American College of Emergency Physicians, 2004. Available from: http://www.acep.org/WorkArea/DownloadAsset.aspx?id=8872 (last accessed: April 2012).
[13] M. Janssen, T. van Achterberg, M. Adriaansen, C.S. Kampshoff, J. Mintjes-de Groot, Adherence to the guideline 'triage in emergency departments': a survey of dutch emergency departments, J. Clin. Nurs. 20 (17–18) (2011) 2458–2468.
[14] P.C. Sprivulis, J. Da Silva, I.G. Jacobs, A.R. Frazer, G.A. Jelinek, The association between hospital overcrowding and mortality among patients admitted via western Australian emergency departments, Med. J. Aust. 184 (5) (2006) 208–212.
[15] J.M. Pines, J.A. Hilton, E.J. Weber, A.J. Alkemade, H. Al Shabanah, P.D. Anderson, M. Bernhar, et al., International perspectives on emergency department crowding, Acad. Emerg. Med. 18 (12) (2011) 1358–1370.
[16] J.M. Pines, R.L. Pilgrim, S.M. Schneider, B. Siegel, P. Viccellio, Practical implications of implementing emergency department crowding interventions: summary of a moderated panel, Acad. Emerg. Med. 18 (12) (2011) 1278–1282.
[17] N. Farrokhnia, K.E. Göransson, Swedish emergency department triage and interventions for improved patient flows: a national update, Scand. J. Trauma Resusc. Emerg. Med. 19 (1) (2011) 72–77.
[18] S.L. Bernstein, D. Aronsky, R. Duseja, S. Epstein, D. Handel, U. Hwang, M. McCarthy, et al., The effect of emergency department crowding on clinically oriented outcomes, Acad. Emerg. Med. 16 (1) (2009) 1–10.
[19] D.J. Magid, B.R. Asplin, R.L. Wears, The quality gap: searching for the consequences of emergency department crowding, Ann. Emerg. Med. 44 (6) (2004) 586–588.
[20] J.R. Richards, M.L. Navarro, R.W. Derlet, Survey of directors of emergency departments in California on overcrowding, West. J. Med. 172 (6) (2000) 385–388.
[21] D.P. Andrulis, A. Kellermann, E.A. Hintz, Emergency departments and crowding in United States teaching hospitals, Ann. Emerg. Med. 20 (1991) 980–986.
[22] S. Liu, C. Hobgood, J.H. Brice, Impact of critical bed status on emergency department patient flow and overcrowding, Acad. Emerg. Med. 10 (4) (2003) 382–385.
[23] K.V. Rondeau, L.H. Francescutti, J. Zanardelli, Emergency department overcrowding: the impact of resource scarcity on physician job satisfaction, J. Healthcare Manage. 50 (5) (2005) 327–340.
[24] L.A. Jarousse, ED throughput: a key to patient safety, Hosp. Health Networks 85 (8) (2011) 33.
[25] M. McCarthy, Overcrowding in emergency departments and adverse outcomes, Br. Med. J. 6 (342) (2011) d2830.
[26] A. Guttmann, M. Schull, M. Vermeulen, T. Stukel, Association between waiting times and short term mortality and hospital admission after departure from emergency department: population based cohort study from Ontario, Canada, Br. Med. J. 342 (2011) 1250–1258.
[27] J. Jensen, Going with the flow. Tracking system helps midwest hospital streamline patient flow and lower emergency room divert rate, Health Manag. Technol. 24 (12) (2003) 43–45.
[28] T. Falvo, L. Grove, R. Stachura, W. Zirkin, The financial impact of ambulance diversions and patient elopements, Acad. Emerg. Med. 14 (1) (2007) 58–62.
[29] B.R. Holroyd, M.J. Bullard, K. Latoszek, D. Gordon, S. Allen, S. Tam, S. Blitz, P. Yoon, B.H. Rowe, Impact of a triage liaison physician on emergency department overcrowding and throughput: a randomized controlled trial, Acad. Emerg. Med. 14 (8) (2007) 702–708.
[30] J. Zuba, Finance Department, Saint Vincent Hospital, Worcester MA, USA. Personal Correspondence.
[31] G. FitzGerald, G.A. Jelinek, D. Scott, M.F. Gerdtz, Emergency department triage revisited, Emerg. Med J. 27 (2010) 86–92.
[32] N. Gilboy, D. Travers, R. Wuerz, Re-evaluating triage in the new millennium: a comprehensive look at the need for standardization and quality, J. Emerg. Nursing. 25 (6) (1999) 468–473.
[33] D. O'Brien, A. Williams, K. Blondell, G.A. Jelinek, Impact of streaming "fast track" emergency department patients, Aust. Health Rev. 30 (4) (2006) 525.
[34] The Advisory Board Company, the High Performance ED: Practice #8: Split-Flow Management. Available at: http://www.advisoryboardcompany.com/ (last accessed: June 2012).
[35] C. DeFlitch, D. Eitel, G. Geeting, R. Cherukuri, M. Escott, J. Smith, J. Armold, S. Paul, Physician-directed queuing (PDQ) improves health care delivery in the ED: early results, Ann. Emerg. Med. 50 (3) (2007) 125–126.
[36] J.L. Wiler, C. Gentle, M. Halfpenny, J.M. Hein, A. Mehrotra, M.G. Mikhail, D. Fite, Optimizing emergency department front-end operations, Ann. Emerg. Med. 55 (2) (2010) 142–160.
[37] T.C. Chan, J.P. Killeen, D. Kelly, D.A. Guss, Impact of rapid entry and accelerated care at triage on reducing emergency department patient wait times, lengths of stay, and rate of left without being seen, Ann. Emerg. Med. 46 (6) (2005) 491–497.
[38] R.S. Valdez, E. Ramly, P.F. Brennan, Industrial and Systems Engineering and Health Care: Critical Areas of Research—Final Report, in: AHRQ Publication no. 10-0079, Agency for Healthcare Research and Quality, Rockville, MD, 2010 (prepared under contract no. 290–09-00027U).
[39] M. Günal, M. Pidd, Discrete event simulation for performance modelling in health care: a review of the literature, J. Simul. 4 (1) (2010) 42–51.
[40] S.A. Paul, M.C. Reddy, C.J. DeFlitch, A systematic review of simulation studies investigating emergency department overcrowding, Simulation 86 (8–9) (2010) 559–571.
[41] J.L. Wiler, R.T. Griffey, T. Olsen, Review of modeling approaches for emergency department patient flow and crowding research, Acad. Emerg. Med. 18 (12) (2011) 1371–1379.
[42] S. Takakuwa, H. Shiozaki, Functional analysis for operating emergency department of a general hospital, in: Proc 2004 Winter Simul Conf., 2004, pp. 2003–2011.
[43] R.E. Blasak, D.W. Starks, W.S. Armel, M.C. Hayduk, Healthcare process analysis: the use of simulation to evaluate hospital operations between the emergency department and a medical telemetry unit, in: Proc 2003 Winter Simul Conf., 2003, pp. 1887–1893.
[44] A. Komashie, A. Mousavi, Modeling emergency departments using discrete event simulation techniques, in: Proc 2005 Winter Simul Conf., 2005, pp. 2681–2685.
[45] S. Samaha, W.S. Armel, D.W. Starks, Emergency departments: the use of simulation to reduce the length of stay in an emergency department, in: Proc 2003 Winter Simul Conf., 2003, pp. 1907–1911.
[46] G.W. Evans, T.B. Gor, E. Unger, A simulation model for evaluating personnel schedules in a hospital emergency department, in: Proc 1996 Winter Simul Conf., 1996, pp. 1205–1209.
[47] T.D. Clark, C.W. Waring, A simulation approach to analysis of emergency services and trauma center management, in: Proc 1987 Winter Simul Conf., 1987, pp. 925–931.
[48] B.A. Tan, A. Gubaras, N. Phojanamongkolkij, Schedule evaluation: simulation study of Dreyer urgent care facility, in: Proc 2002 Winter Simul Conf., 2002, pp. 1922–1927.
[49] T. Coats, S. Michalis, Mathematical modelling of patient flow through an accident and emergency department, Emerg. Med. J. 8 (3) (2001) 190–192.
[50] M.D. Rossetti, G.F. Trzcinski, S.A. Syverud, Emergency department simulation and determination of optimal attending physician staffing schedules, in: Proc 1999 Winter Simul Conf., 1999, pp. 1532–1540.
[51] A. Kumar, R. Kapur, Discrete simulation application-scheduling staff for the emergency room, in: Proc 1989 Winter Simul Conf., 1989, pp. 1112–1120.
[52] J.C. Lowery, Multi-hospital validation of critical care simulation model, in: Proc 1993 Winter Simul Conf., 1993, pp. 1207–1215.
[53] J.C. Lowery, Simulation of a hospital's surgical suite and critical care area, in: Proc 1992 Winter Simul Conf., 1992, pp. 1071–1078.
[54] S. Elbeyli, P. Krishnan, In-patient flow analysis using ProModelTM simulation package, FREC Staff Paper 3–6, University of Delaware, 2000.
[55] D.C. Lane, C. Monefeldt, J.V. Rosenhead, Looking in the wrong place for healthcare improvements: a system dynamics study of an accident and emergency department, J. Oper. Res. Soc. 51 (5) (2001) 518–531.
[56] C. Duguay, F. Chetouane, Modeling and improving emergency department systems using discrete event simulation, Simulation 3 (4) (2007) 311–320.
[57] D.M. Ferrin, M.J. Miller, D.L. McBroom, Maximizing hospital financial impact and emergency department throughput with simulation, in: Proc 2007 Winter Simul Conf., 2007, pp. 1566–1573.
[58] M.J. Miller, D.M. Ferrin, M.G. Messer, Fixing the emergency department: a transformational journey with EDSIM, in: Proc 2004 Winter Simul Conf., 2004, pp. 1988–1993.
[59] M.J. Miller, D.M. Ferrin, J.M. Szymanski, Simulating six sigma improvement ideas for a hospital emergency department, in: Proc 2003 Winter Simul Conf., 2003, pp. 1926–1929.
[60] S. Mahapatra, C. Koelling, L. Patvivatsiri, B. Fraticelli, D. Eitel, L. Grove, Emergency departments II: pairing emergency severity index5-level triage data with computer aided system design to improve emergency department access and throughput, in: Proc 2003 Winter Simul Conf., 2003, pp. 1917–1925.
[61] M.A. Bardi, J. Hollingsworth, A simulation model for scheduling in the emergency room, Int. J. Oper. Prod. Manage. 13 (3) (1993) 13–24.
[62] F.F. Baesler, H.E. Jahnsen, M. DaCosta, Emergency departments: the use of simulation and design of experiments for estimating maximum capacity in an emergency room, in: Proc 2003 Winter Simul Conf., 2003, pp. 1903–1906.
[63] M. Kraitsik, A. Bossmeyer, Simulation applied to planning an emergency department expansion, in: Proc 1993 SCS Western Multiconference on Simulation: Simulation in Health Sciences and Services, 1993, pp. 19–27.
[64] J.T. Blake, M.W. Carter, S. Richardson, An analysis of emergency room wait time issues via computer simulation, Inf. Syst. Oper. Res. 34 (4) (1996) 263–273.
[65] D. Sinreich, Y. Marmor, Emergency department operations: the basis for developing a simulation tool, IIE Trans. 37 (3) (2005) 233–245.
[66] B.G. Sobolev, V. Sanchez, C. Vasilakis, Systematic review of the use of computer simulation modeling of patient flow in surgical care, J. Med. Syst. 35 (1) (2011) 1–16.
[67] J.C. Lowery, B. Hakes, L. Keller, W.R. Lilegdon, K. Mabrouk, F. McGuire, Barriers to implementing simulation in health care, in: Proc 1994 Winter Simul Conf., 1994, pp. 868–875.

[68] J.H. Han, D.J. France, S.R. Levin, I.D. Jones, A.B. Storrow, D. Aronsky, The effect of physician triage on emergency department length of stay, J. Emerg. Med. 39 (2) (2010) 227–233.
[69] S.K. Polevoi, J.V. Quinn, N.R. Kramer, Factors associated with patients who leave without being seen, Acad. Emerg. Med. 12 (3) (2005) 232–236.
[70] B.H. Rowe, P. Channan, M. Bullard, S. Blitz, L.D. Saunders, R.J. Rosychuk, H. Lari, W.R. Craig, B.R. Holroyd, Characteristics of patients who leave emergency departments without being seen, Acad. Emerg. Med. 13 (8) (2006) 848–852.
[71] S.J. Weiss, A.A. Ernst, R. Derlet, R. King, A. Bair, T.G. Nick, Relationship between the national ED overcrowding scale and the number of patients who leave without being seen in an academic ED, Am. J. Emerg. Med. 23 (3) (2005) 288–294.
[72] J. Banks, J.S. Carson, B.L. Nelson, Discrete-Event System Simulation, fourth ed., Pearson Education Inc., Toronto, Canada, 2006.
[73] A.M. Law, Simulation and Modeling Analysis, fourth ed., McGraw-Hill Companies, Inc., Toronto, Canada, 2007.
[74] J. Considine, M. Kropman, H.E. Stergiou, Effect of clinician designation on emergency department fast track performance, Acad. Emerg. Med. 27 (11) (2010) 838–842.
[75] M.S. McClelland, D. Lazar, V. Sears, M. Wilson, B. Siegel, J.M. Pines, The past, present, and future of urgent matters: lessons learned from a decade of emergency department flow improvement, Acad. Emerg. Med. 18 (12) (2011) 1392–1399.
[76] S. Schneider, F. Zwemer, A. Doniger, R. Dick, T. Czapranski, E. Davis, Rochester, New York—a decade of emergency department overcrowding, Acad. Emerg. Med. 8 (11) (2001) 1044–1050.
[77] A.J. Forster, I. Stiell, G. Wells, A.J. Lee, C. Van Walraven, The effect of hospital occupancy on emergency department length of stay and patient disposition, Acad. Emerg. Med. 10 (2) (2003) 127–133.